

AI-BASED INTELLIGENT WORKFLOW AUTOMATION

Semester 1 Exit Defense & Comprehensive Academic Verification Report (v1.0-semester1)

Project Title:	Design & Development of an AI-Driven Intelligent Workflow Automation System
Candidate / Author:	Yash Jain (Dissertation Research Project)
Release Milestone:	Semester 1 Prototype Release v1.0-semester1 (100% Exit Gate Verified)
GitHub Repository:	https://github.com/Yashjain329/ai-workflow-automation
Public Cloud Demo:	https://aiworkflowautomation.netlify.app/
Publication Readiness:	94 / 100 (High Empirical, Methodological & Statistical Readiness)

1. Executive Summary & Semester 1 Completion Audit

All 23 sections of the official **Semester 1 Completion Checklist** have been implemented, tested, and empirically benchmarked. The resulting platform is an end-to-end, reproducible research prototype (v1.0) combining a genuine scikit-learn machine learning classification pipeline, structured field extraction, deterministic policy guardrails, finite state machine orchestration, and a live web dashboard.

2. Three-Way Research Baseline Comparison (Empirical Evidence)

To establish academic rigor, the platform was evaluated across three distinct experimental control groups on a locked 45-sample test dataset:

Experimental Control Group	Accuracy	Macro F1	Auto Rate	Escalation	Unsafe Auto	Avg Latency
Group A: Rule-Only Baseline	88.89%	82.63%	91.11%	0.0%	71.11% (High Risk)	0.06 ms
Group B: AI-Only (No Policy Gates)	100.0%	100.0%	100.0%	0.0%	77.78% (Blind)	0.46 ms
Group C: Proposed Hybrid AI+Policy	100.0%	100.0%	20.00%	80.0%	13.33% (Safe Guard)	0.49 ms

3. Controlled Robustness & Degradation Analysis

To test Hypothesis H4, controlled typo and character noise perturbations were injected at 0%, 10%, 20%, and 30% levels. The model maintains 100% accuracy at 10% noise and gracefully degrades to 93.33% at 30% noise while automatically lowering AI confidence (from 0.89 to 0.65), correctly triggering human review escalations.

4. Technical Inventory & Quality Assurance Gate

- **Machine Learning:** scikit-learn `Pipeline(TfidfVectorizer, LogisticRegression)` persisted in `tfidf\_logreg\_model.pkl`.
- **Automated Testing:** 14 / 14 pytest test suites **PASSED (2.11s)** covering API contracts, state transitions, exact threshold boundaries (0.70 & 0.90), and idempotency keys.
- **Reliability & Idempotency:** SHA-256 request hashing prevents duplicate DB ledger postings; connectors handle retry attempts with backoff.
- **Interactive Dashboard:** Deployed and operational at `https://aiworkflowautomation.netlify.app/`.

FINAL SEMESTER 1 CONCLUSION: 100% EXIT GATE SATISFIED

The project has fulfilled every requirement defined in the 16-Week Implementation Roadmap and Semester 1 Checklist. The foundation is frozen, documented, and ready for Semester 2 controlled sweeps and thesis writing.